

Duality and non-logarithmic nonuniqueness for a multivariable alternative-mean equation

Candidate partial result for arXiv:2605.16876

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Status: candidate substantial partial result, likely valid. Global uniqueness is invariant under $g(x) \mapsto -g(x^{-1})$, giving the new unique endpoint $g(x) = 1 - x^{-1}$. Conversely, an explicit non-logarithmic operator-monotone g has a certified 2×2 three-input example with two solutions. The full classification remains open.

1 Source and open problem

The source is Sejong Kim and Vatsalkumar N. Mer, *A multivariable mean equation arising from the spectral geometric mean*, arXiv:2605.16876 (2026) [1]. Let $\mathbb{P}_n$ be the positive definite $n \times n$ matrices, let $\#$ denote the metric geometric mean, and let

$$\mathcal{L} = \{g : (0, \infty) \rightarrow \mathbb{R} : g \text{ is operator monotone, } g(1) = 0, g'(1) = 1\}.$$

For a positive probability vector $\omega = (w_1, \dots, w_m)$ and $\mathbb{A} = (A_1, \dots, A_m) \in \mathbb{P}_n^m$, define

$$\Gamma_g(\omega; \mathbb{A}) = \left\{ X \in \mathbb{P}_n : \sum_{i=1}^m w_i g(A_i \# X^{-1}) = 0 \right\}. \quad (1)$$

The open problem asks for necessary and sufficient conditions on $g \in \mathcal{L}$ under which this solution is unique. The source records uniqueness for $g(x) = x - 1$, where the solution is the Wasserstein mean, and nonuniqueness for $g(x) = \log x$.

2 Proof intuition

Inversion supplies an exact symmetry of the problem: replacing g by $-g(1/x)$, inverting every datum, and inverting the unknown leaves the equation unchanged. This immediately transfers the Wasserstein uniqueness case to a second endpoint.

For nonuniqueness, deform $\log x$ inside a self-dual operator-monotone family. Self-duality keeps the equation traceless on the determinant-one surface of real 2×2 matrices, so the source's two-dimensional Poincaré–Miranda mechanism survives. One solution is calibrated exactly; strict interval-certified signs on a rectangle force a second one.

3 Duality and a second globally unique endpoint

Call g *globally unique* if $\Gamma_g(\omega; \mathbb{A})$ is a singleton for every $n, m, \omega, \mathbb{A}$.

by the congruence invariance of the generalized Karcher mean. Moreover, from the generalized Karcher equation (4.19), it can be expressed equivalently as

$$\sum_{i=1}^m w_i g(A_i \# X^{-1}) = 0 \quad \text{for some } g \in \mathcal{L}.$$

We denote by $\Gamma_g(\omega; \mathbb{A})$ the set of solutions to the equation $\Phi_g(X) = 0$. By an argument similar to that in Remark 2.2, one can show that $\Gamma_g(\omega; \mathbb{A}) \neq \emptyset$.

Lemma 4.2. *Let $\mathbb{A} = (A_1, A_2, \dots, A_m) \in \mathbb{P}_n^m$ and $\omega = (w_1, w_2, \dots, w_m) \in \Delta_m$ with $\alpha I \leq A_j \leq \beta I$ for all j and some $\alpha, \beta > 0$. Then $X \in \Gamma_g(\omega; \mathbb{A})$ satisfies*

$$\alpha I \leq X \leq \beta I.$$

Theorem 4.3. *Let $\omega = (w_1, w_2, \dots, w_m) \in \Delta_m$. Then there exists $\epsilon_\omega > 1$ such that for any $\mathbb{A} \in [\epsilon_\omega^{-1} I, \epsilon_\omega I]^m \subset \mathbb{P}_n^m$, the equation $\Phi_g(X) = 0$ has a unique solution in $[\epsilon_\omega^{-1} I, \epsilon_\omega I] \subset \mathbb{P}_n$.*

A similar proof can be given for Lemma 4.2 and Theorem 4.3, analogous to the arguments used in Lemma 2.3 and Theorem 2.4. Therefore, we skip the proof.

Open problem: For any $\omega \in \Delta_m$ and $\mathbb{A} = (A_1, \dots, A_m) \in \mathbb{P}_n^m$, we know that the equation (4.21) admits at least one solution. In the particular case $g(x) = x - 1$, the solution is unique and coincides with the Wasserstein mean. In contrast, for $g(x) = \log x$, uniqueness generally fails.

This leads to the following question: determine necessary and sufficient conditions on $g \in \mathcal{L}$ under which the equation (4.21) admits a unique solution.

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Figure 1: Equation (4.21) and the open classification problem on page 21 of the source PDF.

Theorem 1 (Duality). *For $g \in \mathcal{L}$, define $g^\vee(x) = -g(x^{-1})$. Then $g^\vee \in \mathcal{L}$ and*

$$\Gamma_{g^\vee}(\omega; A_1, \dots, A_m) = \{Y^{-1} : Y \in \Gamma_g(\omega; A_1^{-1}, \dots, A_m^{-1})\}. \quad (2)$$

Consequently, g is globally unique if and only if $g^\vee$ is.

Proof. If $0 < A \leq B$, then $A^{-1} \geq B^{-1}$. Operator monotonicity of g , followed by multiplication by -1 , shows $g^\vee(A) \leq g^\vee(B)$. Also $g^\vee(1) = 0$ and $(g^\vee)'(1) = g'(1) = 1$, so $g^\vee \in \mathcal{L}$.

The metric geometric mean is self-dual:

$$(A_i \# X^{-1})^{-1} = A_i^{-1} \# X.$$

Thus X satisfies the $g^\vee$ -equation exactly when

$$0 = - \sum_i w_i g(A_i^{-1} \# X).$$

Putting $Y = X^{-1}$ turns this into the g -equation for the inverse data, proving (2). The uniqueness assertion follows by applying the involution twice. $\square$

Corollary 1. *The function $g_-(x) = 1 - x^{-1}$ is globally unique, and its solution is*

$$\Gamma_{g_-}(\omega; \mathbb{A}) = \{\Omega(\omega; A_1^{-1}, \dots, A_m^{-1})^{-1}\},$$

where Ω denotes the Wasserstein mean.

Proof. The source recalls that $g_+(x) = x - 1$ is globally unique with solution $\Omega(\omega; \mathbb{A})$. Since $g_+^\vee = g_-$, the claim is Theorem 1. $\square$

4 A non-logarithmic family with nonunique equations

For $0 < s \leq 1$, set

$$g_s(x) = \frac{2x^s - 1}{s x^s + 1}, \quad g_0(x) := \log x. \quad (3)$$

Proposition 1. *For $0 < s \leq 1$, $g_s \in \mathcal{L}$, $g_s(x^{-1}) = -g_s(x)$, and $g_s \rightarrow \log$ locally uniformly as $s \downarrow 0$.*

Proof. On the upper half-plane, the principal power $z \mapsto z^s$ again has positive imaginary part. The real Möbius map $z \mapsto (z-1)/(z+1)$ preserves the upper half-plane. Hence g_s is a Pick function, and Löwner's analytic characterization makes it operator monotone [2, Chapter V]. Direct calculation gives $g_s(1) = 0$, $g'_s(1) = 1$, and the self-duality identity. The limit is the scalar expansion $x^s = e^{s \log x}$, uniform on compact subsets. $\square$

Let

$$S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad X_0 = e^{3S}.$$

For a fixed sufficiently small $s > 0$, define

$$\begin{aligned} a_s &= \frac{2}{s} \tanh \frac{3s}{2}, & b_s &= \frac{2}{s} \tanh \frac{19s}{20}, & r_s &= (a_s^2 + b_s^2)^{1/2}, \\ q_s &= \frac{2}{s} \operatorname{artanh} \frac{sr_s}{2}, & Z_1 &= 3S, & Z_2 &= \frac{19}{10}T, \\ Z_3 &= -\frac{q_s}{r_s}(a_s S + b_s T), & B_i &= e^{Z_i}, & A_i &= B_i X_0 B_i. \end{aligned}$$

Here $sr_s/2 < 1$ for small s ; in particular it holds at $s = 1/100$, since $a_s < 3$ and $b_s < 19/10$.

Theorem 2 (Explicit non-logarithmic nonuniqueness). *Let $s = 1/100$, let g_s be given by (3), take equal weights $(1/3, 1/3, 1/3)$, and use the three matrices A_i above. Then $\Gamma_{g_s}(\omega; A_1, A_2, A_3)$ contains at least two distinct matrices. Moreover, there is $\varepsilon > 0$ such that analogous examples are nonunique for every $0 < s < \varepsilon$.*

Proof. The relations $S^2 = T^2 = I$ and $ST = -TS$ give $(a_s S + b_s T)^2 = r_s^2 I$. If U is a self-adjoint involution, functional calculus gives

$$g_s(e^{tU}) = \frac{2}{s} \tanh \left(\frac{st}{2} \right) U.$$

The definition of q_s therefore yields

$$g_s(B_1) = a_s S, \quad g_s(B_2) = b_s T, \quad g_s(B_3) = -a_s S - b_s T.$$

Because $A_i = B_i X_0 B_i$, the Riccati characterization of $\#$ shows $A_i \# X_0^{-1} = B_i$. Hence X_0 is an exact solution.

Now parametrize the real determinant-one positive matrices by

$$X(u, v) = e^{uS+vT}.$$

Every A_i has determinant one, so does $A_i \# X(u, v)^{-1}$. Its two eigenvalues are reciprocal; since $g_s(x^{-1}) = -g_s(x)$, its image under g_s is traceless. Consequently

$$\frac{1}{3} \sum_{i=1}^3 g_s(A_i \# X(u, v)^{-1}) = F_{1,s}(u, v)S + F_{2,s}(u, v)T \quad (4)$$

for continuous real functions $F_{1,s}, F_{2,s}$.

Set

$$R = [1.61247432825, 1.62347432825] \times [0.5194188906, 0.5254188906].$$

Outward-rounded interval arithmetic, subdividing each face into 8,000 exact decimal intervals, certifies at $s = 1/100$:

$$\begin{aligned} F_{1,s}(1.61247432825, v) &\geq 1.5903 \times 10^{-4}, \\ F_{1,s}(1.62347432825, v) &\leq -7.5949 \times 10^{-5}, \\ F_{2,s}(u, 0.5194188906) &\geq 1.1393 \times 10^{-4}, \\ F_{2,s}(u, 0.5254188906) &\leq -2.7880 \times 10^{-5}. \end{aligned}$$

The companion script records the complete enclosures. The Poincaré–Miranda theorem gives $(u_*, v_*) \in R$ with both functions in (4) zero. Thus $X(u_*, v_*)$ is a solution, and it differs from $X_0 = X(3, 0)$.

Finally, the calibrated matrices and all four face functions extend continuously to $s = 0$. The same interval script certifies strict face signs there. Uniform continuity on the four compact faces preserves those signs for every sufficiently small $s > 0$, proving the last assertion. $\square$

5 Verification, limitations, and novelty

The interval script uses the exact determinant-one identity

$$A \# B = \frac{A + B}{\sqrt{\det(A + B)}} \quad (A, B \in \mathbb{P}_2, \det A = \det B = 1)$$

and closed two-by-two functional calculus. It reports zero failed subintervals on every face for both $s = 0$ and $s = 1/100$. The code is a certificate only for the four scalar signs; the duality, operator monotonicity, exact calibration, and topological conclusion are proved above.

The result does not give necessary and sufficient conditions for arbitrary g . It shows that the uniqueness class is closed under duality, adds the endpoint $1 - x^{-1}$, and proves that failure occupies a nontrivial one-parameter neighborhood of the logarithmic mechanism.

The local run indexes were searched for arXiv:2605.16876 and the core alternative-mean and generalized-Karcher terms. Bounded web/arXiv searches on 11 August 2026 used the exact title and question, $1 - x^{-1}$, and the family $(x^s - 1)/(x^s + 1)$. They found the May 2026 source and background literature, but no later classification, no duality theorem for (1), and no matching nonuniqueness family. Novelty confidence is moderate because the source is recent and the search was bounded.

Human review is recommended, focusing on Proposition 1, the factor in q_s , and the interval implementation.

References

- [1] S. Kim and V. N. Mer, *A multivariable mean equation arising from the spectral geometric mean*, arXiv:2605.16876 (2026). <https://arxiv.org/abs/2605.16876>.
- [2] R. Bhatia, *Matrix Analysis*, Graduate Texts in Mathematics 169, Springer, 1997.